[Title]: Longitudinal Gadolinium-Free Contrast-Enhanced T1 Synthesis from a Patient's Prior Exam with a Schrödinger Bridge

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Impact

Synthesizing post-contrast MRI could spare patients repeated gadolinium injections, but the problem is ill-posed: enhancement is not always evident in pre-contrast images. The patient's prior contrast-enhanced exam supplies the missing information, improving synthesis of the follow-up scan.

Synopsis

Motivation: Patients under surveillance for brain metastases receive gadolinium at every visit; their prior post-contrast exam is available but unused by contrast-synthesis methods.
Goal(s): Synthesize the current post-contrast T1 (CT1) from the current non-contrast exam and the prior exam, using only information available at inference.
Approach: An image-to-image Schrödinger bridge (I2SB) maps T1 to CT1, conditioned on FLAIR/T1/T2 and the prior T1/CT1 at five adjacent slices, re-registered from non-contrast contrasts only; [N] patients, 1007 test studies.
Results: The prior exam raised PSNR by 0.94 dB and SSIM by 3.0 points ($p < 10^{-100}$, better in 97% of studies). [tumour region]

Introduction

Brain metastases are monitored with serial gadolinium-enhanced MRI[1], and repeated gadolinium exposure motivates synthesizing post-contrast images from non-contrast ones[2]. Single-exam synthesis must infer enhancement from T1, T2 and FLAIR alone. In surveillance, however, every patient has a prior post-contrast exam, highly correlated with the current one. We condition an image-to-image Schrödinger bridge (I2SB)[3] on that prior exam, and evaluate it under inference-realistic conditions: the prior is registered, and all images are masked, without using the current CT1.

Methods

Data. NYUMets[4]: longitudinal brain-metastasis MRI (FLAIR, T1, CT1, T2) [scanner/protocol]. Studies of a patient were placed on a common grid by a 3-D affine fit during preprocessing; each volume is scaled by its in-brain median, so prior and current share an intensity scale. Training on slices 50–89 of [N] patients; testing on slices 70–109 of 1007 studies from [N] held-out patients with a prior exam.

Model. I2SB bridges the current T1 (x_1) to the current CT1 (x_0) with a conditional U-Net ([23M] parameters, effective batch 64, EMA, 300k steps), conditioned on FLAIR/T1/T2. The longitudinal model additionally receives the prior T1 and CT1 at slices $z-2 \ldots z+2$ as input channels. Sampling: 20 steps.

Inference-time registration and masking. The preprocessing alignment saw the current CT1 (through its brain and field-of-view masks), which inference cannot. Each prior is therefore re-registered to the current study by a 3-D affine (rotation, per-axis scale, shift) maximising the mean masked correlation of FLAIR, T1 and T2 of both exams, and all four prior contrasts are resampled with it. Images and metrics are masked by the current T1 support intersected with the registered prior CT1 support.

Evaluation. PSNR, SSIM and NRMSE within the mask, averaged per study; paired Wilcoxon signed-rank tests over studies. [Tumour-region error from the radiotherapy segmentations.]

Results

Conditioning on the prior exam improved every metric (Figure 2, Table): PSNR $24.79 \pm 0.09 \rightarrow 25.73 \pm 0.10$ dB, SSIM $87.79 \rightarrow 90.83\%$, NRMSE $13.07 \rightarrow 11.97\%$ (all $p < 10^{-160}$; better in 97% of studies, median $+0.84$ dB). The gain grew from $+0.67$ dB at the skull base (slice 70) to $\approx +1.05$ dB above slice 95 (Figure 3). [Tumour slices: gain inside lesions; new and resolved lesions (Figure 4).] [Example cases, Figure 1.]

Discussion

[Where the prior helps: stable anatomy vs. interval change; cases where it hurts.] Limitations: [the no-prior baseline was trained with fixed intensity scaling and on all patients; one network family; slice-wise 2-D.]

Conclusion

A patient's prior exam, registered without the scan being predicted, improves gadolinium-free synthesis of the follow-up post-contrast T1. [one clause on tumours]

Acknowledgements

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References

[1] [RANO-BM / surveillance imaging reference.]

[2] [Gadolinium-free contrast synthesis reference(s).]

[3] Liu GH, Vahdat A, Huang DA, Theodorou EA, Nie W, Anandkumar A. I^2SB: Image-to-Image Schrödinger Bridge. In: Proc ICML. 2023. [check]

[4] [NYUMets dataset reference.]

Figures

figures/make_figure.py figures/configs/nyumets_longi.py (rows picked in the viewer)

Figure 1: Example cases [(rows)]: current T1 (input), registered prior CT1, synthesis without and with the prior exam, and the acquired CT1; magnified region with its 5× residual, PSNR/SSIM per panel.

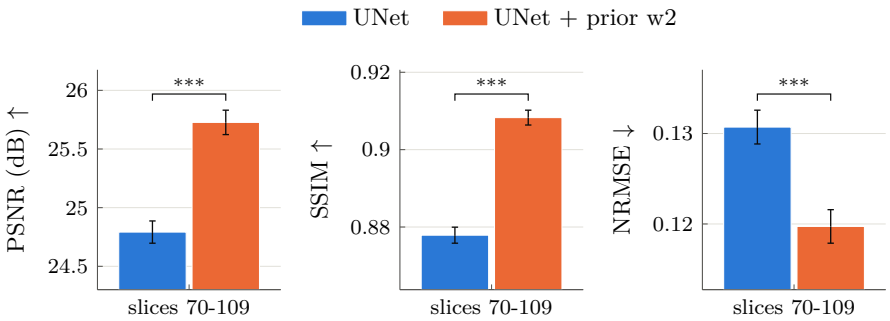
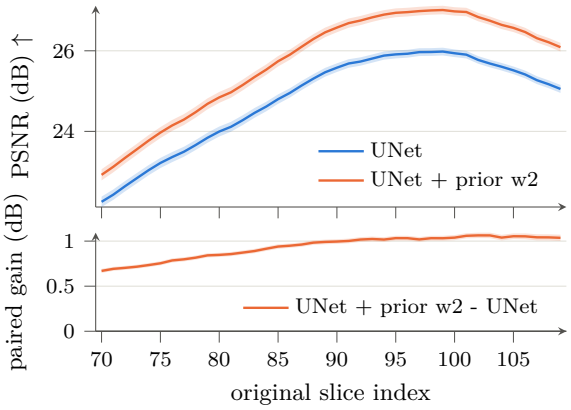


Figure 2: 1007 test studies, slices 70–109: mean ± s.e. over studies; *** paired Wilcoxon $p \leq 0.001$. Value axes do not start at zero.



Method	PSNR (dB) ↑	SSIM (%) ↑	NRMSE (%) ↓
slices 70-109 (n=1007 volumes)			
UNet	$24.79 \pm 0.09^{*}$	$87.79 \pm 0.21^{*}$	$13.07 \pm 0.19^{*}$
UNet + prior w2	$\mathbf{25.73} \pm 0.10$	$\mathbf{90.83} \pm 0.19$	$\mathbf{11.97} \pm 0.19$

Figure 3: Top: PSNR per original slice index (mean ± s.e. over studies) and the paired gain of the prior exam. Bottom: summary; bold best, * $p < 0.05$ vs. the longitudinal model.

Tumour cases: scripts/nyumets_tumor_slices.py –write-rows, then figures/make_figure.py

Figure 4: [Lesion-containing slices: new, stable and resolved lesions; error inside the lesion without and with the prior exam.]